

1 Introduction

Ledrappier's 1999 paper *Distribution des orbites des réseaux sur le plan réel* [ref] proves an ergodic theorem for averages of $\mathrm{SL}(2, \mathbb{R})$ orbits of real-valued functions. The goals of this document are twofold (i) rewrite Ledrappier's paper in the language of Teichmüller theory, to better motivate and understand the result, and (ii) to explicate terse elements of Ledrappier's proof.

The following notation will be used throughout

- $\mathbb{T}^2$ will denote the (topological) torus
- $\mathbf{T}_1$ will denote the Teichmüller space of $\mathbb{T}^2$
- Γ will denote $\mathrm{SL}(2, \mathbb{Z})$, the mapping class group of $\mathbb{T}^2$
- $\mathcal{Q}^1$ will denote the space of unit quadratic differentials on $\mathbb{T}^2$
- $\mathcal{M}$ will denote the moduli space $\mathbf{T}_1/\Gamma$
- Fix once and for all curves α, β with homology classes giving a basis for $\mathbb{T}^2$ and $i(\alpha, \beta) = 1$.

2 Teichmüller Theory for the Torus

Definition 1 *The Teichmüller space of the torus $\mathbf{T}_1$ will be defined as the set of pairs (S, φ) where S is a Riemann surface, and $\varphi : \mathbb{T}^2 \rightarrow S$ is an orientation preserving diffeomorphism, two pairs $(S_1, \varphi_1), (S_2, \varphi_2)$ will be identified if there is a biholomorphism $I : S_1 \rightarrow S_2$, such that $I\varphi_1$ is homotopic to φ_2 .*

In order to work with Teichmüller space, I will identify it with the space of abelian differentials of the form $dx + \tau dy$ for $\tau \in \mathbb{H}^2$. An abelian differential on S is a section of the holomorphic cotangent bundle K_S , on the torus the transition functions are given by translation, $z_\alpha = z_\beta + c$, thus the transition functions $\frac{dz_\alpha}{dz_\beta}$ of K_S are constant, since the torus is compact any such section must be given by a constant function by the maximum principle. Now fix an abelian differential ω , we consider the period coordinates $(A, B) = \left(\int_{\varphi(\alpha)} \omega, \int_{\varphi(\beta)} \omega \right)$, the choice of ω is only defined up to scaling, but we may normalize by replacing ω with $A^{-1}\omega$ to get period coordinates $\left(1, \int_{\varphi(\beta)} \omega \right)$. Now considering the covering $\mathbb{R}^2 \rightarrow S$, we get that up to homotopy $\varphi(\alpha)$ corresponds to the deck transformation $(x, y) \mapsto (x + 1, y)$, and similarly $\varphi(\beta)$ to $(x, y) \mapsto (x, y + 1)$, thus we get a unique representative $\omega = dx + \tau dy$, for $\tau = A^{-1}B$. To see that $\tau \in \mathbb{H}^2$, we note that since φ is orientation preserving the area form is taken to be positive so that

$$0 < \mathrm{Area}(S) = \int_S \frac{i}{2} \omega \wedge \bar{\omega} = [\text{computation here}]$$